

Project Design Phase

Solution Architecture

Date	14 July 2026
Team ID	
Project Name	SHOPEZ: E-commerce Application
Maximum Marks	4 Marks

Solution Architecture

The Solution Architecture of ShopEZ – MERN E-Commerce Platform defines how the frontend, backend, database, and external services interact to provide a secure and efficient online shopping experience. The application follows the MERN architecture, where the React frontend communicates with the Node.js and Express.js backend through REST APIs. The backend processes user requests, performs business logic, and stores or retrieves data from MongoDB. Firebase Authentication is used for secure user registration and login, while external services such as Cloudinary can be used for image storage and email services for order notifications. This architecture ensures scalability, security, maintainability, and smooth communication between all system components.

Goals:

- Find an efficient technology solution for online shopping.
- Define the interaction between the frontend, backend, database, and external services.
- Ensure secure authentication and reliable data management.
- Provide a scalable, maintainable, and user-friendly e-commerce platform.

Solution Architecture

